SCIENCE

- From the Latin word "scientia," which means "knowledge"
- Any methodological activity, such as observation, experimental investigation, and theoretical explanation, aimed at discussing natural phenomena
- Systematized knowledge based on facts

Divisions of Science

- Social Science
- Political Science
- History
- Mathematics
- Natural science – deals with nature and natural phenomena

Branches of Natural Science

- **Physical Science** – deals with non-living things
- **Biological Science** – deals with living things

Scientific Method

- An orderly, logical, and rational manner of solving problems
- Enables and leads scientists toward unveiling the truths about observable phenomena, helping them construct their clear representations
- **Steps in Solving Problems**
    - **The Problem** – question formulated from observations to which you want to find specific answers
        - Identify, state, know, and understand the problem
        - Identify the variables

          Types of variables
            - Independent variable – the one which is manipulated or changed
            - Dependent variable – the one which is measured and observed; depends on the independent variable

          Example: In the experiment, "the effect of sunlight on the growth of the plants"

          Sunlight - independent variable
          Growth - dependent variable
    - **Hypothesis**
        - It is an inference or a scientific guess.
        - It can be a positive or a negative hypothesis.
    - **Experimentation**
        - Designed and conducted to test the validity of the hypothesis
        - A good experiment is one where the least time, effort, and logistics are used, but yields the best result.

- Steps in Performing an Experiment
  - Gather the materials to be used in experimentation
  - Set the materials
  - Observe
  - Gather the data
  - Interpret the data

- **Conclusion**
  - Draw the conclusion by stating the findings in the manner that allows the problem to be solved / answered.
- **Application**
  - This is an optional step. If the results of the experiment are good enough, then there will be a need to apply the results of the experiment in real-life situations.

Scientific Attitude

- This is the scientist's way of doing and thinking, especially when performing an experiment or any scientific investigation.
- Some of the scientific attitudes are the following:
  - **Curiosity** – pays particular interest and asks questions about observations
  - **Honesty** – reports results of observations truthfully
  - **Humility** – accepts that he does not have the monopoly of ideas about the phenomenon
  - **Unbiased** – ability to separate his own personal ideas from the actual findings
  - **Open-mindedness** – readiness to accept or consider opinions of others
  - **Patience** and **perseverance** – does not easily give up when investigation seems difficult

Law vs. Theory

- **Law** – generalizations formulated from observations that are proven correct and true.
  Ex. Ohm's Law: that current is directly proportional to the voltage but inversely proportional to the resistance. This explains that an increase in voltage will increase the current if the resistance is constant and if the voltage is held constant, and increase in resistance will give a decrease in current.
- **Theory** – generally accepted explanations of observations, but are yet to be fully proven because some questions still need to be answered.
  Ex. Theory of Evolution: results of scientific investigations explain changes in organisms, but questions that cannot undergo experimentations are left unanswered.

PHYSICAL SCIENCES

Branches of Physical Science

- **Physics** – deals with matter and energy and of the interactions between the two
- **Chemistry** – deals with the composition and properties of matter
- **Earth Science** – deals with the physical aspects of the Earth

Physics

Scientific Notation

- Short-hand in writing extremely large or small numbers using the power of ten notation
- Takes the form A x 10^n, where A is an integer from 1 to 9.9999 and n is the exponent in the power of ten
- The exponent n may be dealt also as the number of times the decimal point is moved: positive if moved to the left, and negative when moved to the right.

Ex. 361000 = 3.61 x 10^5 - decimal point is moved to the left 5 times so that the number to the left of the decimal point is less than ten.

Ex. 000361 = 3.61 x 10^{-4} - decimal point is moved to the right 4 times so that the number to the right of the decimal point is less than ten.

Measurement

- The process of comparing the quantity to be measured and the corresponding standard
- The quantity is the one being measured, not the object

Physical Quantities

- There are two physical quantities, namely: the fundamental quantities and the derived quantities.
- The fundamental quantities are also called basic quantities. The following are the fundamental quantities, together with their corresponding standard units:

▪ length	meter
▪ mass	kilogram
▪ time	second
▪ temperature	Kelvin
▪ luminous intensity	candela
▪ electric current	ampere

- Derived quantities are quantities taken from fundamental quantities.

  Ex. Area = l x l, area is taken from the fundamental quantity, length

  Density = mass/volume, and volume = l x l x l, density is derived from fundamental quantities length and mass.

- **Relationship between Quantities**
  - **Direct Proportionality** – When one quantity increases, the other quantity increases also in proportion to each other.
    - A is directly proportional to B
    - A/B = K
    - If A is plotted against B, the resulting graph line is straight line slanting to the right
  - **Direct-square Proportionality** – When one quantity increases, the other quantity increases also, but faster and greater than the other.

- is directly proportional to B2
- A/B2 = K
- If A is plotted against B, the resulting graph line is a parabola.

- **Inverse Proportionality** – When one quantity increases, the other quantity decreases.
  - A is inversely proportional to B.
  - AB = K
  - If A is plotted against B, the resulting graph line will be a hyperbola.

- Physical quantities are also classified as vector and scalar quantities.
  - Scalar quantities are quantities with magnitude only.
    Ex. Area, density, distance, speed
  - Vector quantities are quantities with both magnitude and direction.
    Ex. Displacement, velocity, force
    - Vector quantities can be represented by an arrow, the tail represents the magnitude and the arrowhead represents the direction.

Mechanics

- **Force**
  - Push or pull that changes or tends to change the motion of the body
  - It is a vector quantity.
  - Its unit is newton or kg-m/sec2 in MKS.
  - Sets of forces
    - **Parallel forces** – forces whose lines of action are parallel to each other
    - **Concurrent forces** – forces whose lines of action meet at a common point
  - **Friction** – a force between two surfaces that opposes the motion of the body
  - **Laws of Equilibrium**
    - First law of equilibrium – for a body to be at rest, the sum of all the forces should be equal to zero.
    - Second law of equilibrium – for a body to be at rest, the sum of all the torques should be equal to zero.
- **Motion**
  - Continuous change in position of a body with respect to a reference point
  - **Speed** is the rate of motion.
    v = d/t, where v = speed, d = distance, and t = time
  - **Velocity** is also the rate of motion, only it is a vector.
  - **Uniform motion** is one where the speed or velocity remains constant.
  - **Acceleration** is the change in velocity at a given time interval.
  - **Rectilinear motion** – motion along a straight path. An example of this is the motion of a freely-falling body.
  - **Curvilinear motion** – motion along a curved path. Circular motion and projectile motion are examples of curvilinear motion.
  - **Newton's Three Laws of Motion**
    - **Law of Inertia** – a body at rest will remain at rest and a body in motion will

continue to move in uniform motion unless acted upon by an unbalanced force.

- **Law of Acceleration** – When acted upon by a net force, the body will accelerate, and that the acceleration of the body is directly proportional to the force applied and inversely proportional to the mass of the body. $F = ma$.
- **Law of Interaction** – for every action, there is an equal but opposite reaction.

- **Work**
  - The product of the force applied and the displacement through which the force is directed.
    $W = Fd$, measured in N.m or joule (J)
  - Can be done easier and faster with the use of machines
    - **Machine** is any device that helps in doing work.
    - The six **simple machines** are: lever, pulley, wheel, and axle, inclined plane, screw, and wedge.
    - Mechanical advantage is the number of times a machine can do work compared to the work done without a machine.
    - The efficiency of a machine is the ratio of work output to work input.
      Efficiency = work output/work input x 100%
- **Power**
  - Rate of doing work
    $P = W/t$, measured in joule/sec or watt
- **Energy**
  - Ability of doing work
  - **Forms of energy**
    - **Mechanical energy**
      - **Potential energy** (PE) – energy due to position of the body
        $PE = mgh$, where m = mass of the body, h = height and g = acceleration due to gravity which is 9.8 m/s^2.
      - **Kinetic energy** (KE) – energy of the body which is in motion
        $KE = 1/2mv^2$
    - **Electrical energy** – energy of the moving electrons
    - **Chemical energy** – energy in the atoms and molecules of matter
    - **Nuclear energy** – energy resulting from the fission or fusion reactions in atoms
    - **Radiant energy** – energy in the form of electromagnetic waves
    - **Thermal energy** – internal energy of a body which is the total kinetic energy of the molecules of a body

Fluids

- **Density** is the mass of the substance at a given volume, $d = m/V$.
- If the density of the material or substance is greater than the density of the liquid, the material will sink in the liquid; if the density of the material or substance is less than the liquid, the material will float in the liquid.
- The greater the density of the liquid, the greater is the buoyant force.
- **Pressure** is the force exerted at a given unit area, $p = F/A$

- **Liquid pressure** depends on the density and height of the liquid. Liquid pressure, p = dhg where p is the liquid pressure, d is the density of the liquid, and h is the height of the liquid. g is a constant and does not affect the pressure.
- **Atmospheric pressure** or **air pressure** depends on the height. The higher the position, the thinner is the air, and the lesser is the air pressure.

Heat and Temperature

- **Heat** is the total kinetic energy of molecules of a body.
- **Temperature** is the average kinetic energy of molecules.
- **Temperature conversion**
  - oC = 5/9 (oF – 32o)
  - oF = 9/5 C + 32o
  - oK = oC + 273o
- **Methods of Heat Transfer**
  - **Conduction** – heat transfer from one molecule to adjacent molecule
  - **Convection** – heat transfer by the actual movement of heated molecules. The circulation of the heated molecules is called **convection current**.
  - **Radiation** – transfer of heat across space in the form of electromagnetic waves
- **Change of Phase**
  - **Melting or Fusion** – change of phase from solid to liquid at its normal melting point
  - **Freezing or Solidification** – change of phase from liquid to solid at its normal freezing point
  - **Vaporization** – change of phase from liquid to gas at its normal boiling point
  - **Condensation** – change of phase from gas to liquid
  - **Sublimation** – change of phase from solid to gas without passing the liquid state

Wave and Sound

- A **wave** is a disturbance propagated through a medium. There are two kinds of waves: transverse wave and longitudinal wave.
- The properties of waves are reflection, refraction, diffraction, and interference.
- Sound is a longitudinal wave, coming from a vibrating source, transmitted through a medium and interpreted by the sense of hearing.

Light

- Theories about the nature of light:
  - **Corpuscular Theory** – Light is made up of particles known as corpuscles.
  - **Wave Theory** – Light is a wave.
  - **Electromagnetic Wave Theory** – Light is composed of electric field and magnetic field.
- Since light behaves as a wave, it can be reflected, refracted, diffracted, and interfered.
- The image formed by a mirror is due to reflection of light. The kinds of mirror are plane and curved mirrors. Curved mirrors can be convex or concave.
- The image formed by a lens is due to refraction of light. Lenses are categorized as diverging and converging.

Electricity

- **Classifications of electricity:**
  - **Electrostatics** is electricity at rest. It only involves electric charges and their behaviour. Electric charges are of two kinds:
    - Negative charge – when there is an excess of electrons
    - Positive charge – when there is a deficiency of electrons
  - **Current electricity** – electricity due to moving charges
  - **Basic components of electric circuit:**

Quantity	Unit	Measuring Device
Current	Ampere	Ammeter
Voltage	Volt	Voltmeter
Resistance	Ohm	Ohmmeter

  - Ohm's Law states that the current flowing is directly proportional to voltage and inversely proportional to resistance. $I = V/R$.

Electromagnetism

- **Magnetism** – derived from Magnesia, an island in the Aegean Sea
  - **Magnet** – an object that attracts magnetic objects like metals
  - A magnet has two poles, north and south.
  - Like poles repel; unlike poles attract.
- **Generator** – a device that changes mechanical energy to electrical energy
- **Motor** – a device that changes electrical energy to mechanical energy

Nuclear Energy

- **Atom** – The smallest and indivisible particle of matter it is composed of sub-particles, namely: **electron**, which carries negative charge, **proton** which is positively charged; and **neutron**, which is neutral.
- **Types of Nuclear Reaction:**
  - **Fission** – when a heavy nucleus splits together with the release of energy
  - **Fusion** – when two light nuclei combine with the release of energy

Chemistry

- Study of the composition and changes in matter

Matter

- Anything that has mass and has volume
  Mass – amount of the substance indicative of the inertia it possesses
  Volume – space occupied
- **Classification of Matter:**
  - **Pure substances** – made of definite kind of material
    - **Elements** – made of the same atoms
      Ex. Gold, aluminium, oxygen, carbon, platinum, etc.
    - **Compounds** – made of 2 or more different atoms in definite composition or

ratio and which cannot be separated by ordinary physical means
Ex. Water – H_2O – composed of 2 atoms of hydrogen and 1 atom of oxygen
Glucose - $C_6H_{12}O_6$

- **Mixtures** – physical combination of 2 or more substances which can be separated by mechanical means
  - **Homogeneous mixture** – a mixture in which the molecules are thoroughly mixed; a mixture that is uniform throughout.
    - Solutions are homogeneous mixtures. The components of solution are **solute**, which is the dissolved particles, and the **solvent**, which is the dissolving particles.
  - **Heterogeneous mixture**
    - **Suspensions** – heterogeneous mixture where particles are too large that they settle at the bottom of the container
    - **Colloids** – heterogeneous mixtures whose particles are not large enough to settle nor small enough to be dissolved, like the Tyndall Effect, which is the scattering of light by the particles
- **Methods of Separating Mixtures**
  - **Filtration** – use of filter paper to separate liquid from solid components
    Filtrate – liquid that passes through the filter paper
    Residue – substance that did not pass through the filter paper
  - **Decantation** – pouring off a layer of liquid from a mixture
  - **Magnetism** – use of magnets to separate magnetic materials from the nonmagnetic ones
  - **Centrifugation** – Substance is subjected to circular or rotational motion in a centrifuge.
  - **Distillation** – Liquid is set to boiling. Vapour is collected and later cooled in order to condense.
  - **Chromatography** - passing mixture in solution or suspension or as a vapour (as in gas chromatography) through a medium in which the components move at different rates

- **Properties of Matter**
  - **Physical properties**
    - **Extensive** – depends on the amount of the substance
      Ex. mass, volume, weight, pressure
    - **Intensive** – does not depend on the amount of substance
      Ex. density, taste, colour, smell, specific heat capacity, melting or boiling point, malleability, etc.
  - **Chemical Properties**
    - Property exhibited due to composition of the substance
      Ex. combustion, corrosion, decomposition, etc.
- **Changes in Matter**
  - **Physical change** – change in form and appearance but not in the composition
    - **Change in phase**: melting, evaporation, condensation, sublimation, freezing

- Change in form or shape: cutting, breaking, folding, etc

- **Chemical change** – change in the composition resulting in the formation of a new substance
  - Observed when substances react with each other and form a different substance.
  - Forming precipitates, evolution of gas, release or absorption of heat, change in color and taste, production of sound and light
- **Types of Chemical Reactions**
  - **Combination or synthesis** – 2 simple substances combine to form one new substance

    A + B → AB

    2H2 + O2 → 2H2O
  - **Decomposition Reaction** – a substance breaks down into simpler substances

    AB → A + B

    2H2O → 2 H_2 + O_2
  - **Single Replacement Reaction** – a more active element replaces a less active atom in a compound

    A + BC → AC + B

    2K + ZnCl2 → 2 KCl + Zn

    2KCl + Zn → No Reaction because Zn cannot replace K which is more active
  - **Double Replacement Reaction** (Metathesis) – replacement of the ions

    AB + CD → AD + CB

    2NaCl + K_2S → Na_2S + 2 KCl
  - **Neutralization** – reaction between an acid and a base resulting in the formation of salt and water

    H_2SO_4 + $Mg(OH)_2$ → $MgSO_4$ + 2 H_2O

    Acid + base → salt + water

- **Law of Conservation of Energy**
  - During chemical reaction, the energy before and after the reaction is constant. Energy is not created nor destroyed, but only transformed.

- **Law of Conservation of Mass**

  Total mass of the reactants before the reaction is equal to the total mass of the products after the reaction.

- **Phases of Matter**
  - **Solid** – molecules are compact; small intermolecular spaces; strong molecular attraction; molecular motion limited to vibrations; definite shape, size and volume
  - **Liquid** – molecules are farther apart than solids, larger intermolecular spaces; weaker intermolecular force of attraction; molecules move pass one another, definite volume; takes the shape of container
  - **Gas** – molecules are far apart; large intermolecular spaces; very weak intermolecular force of attraction; molecules are free to move; no definite volume; takes the shape of the container

Kinetic Molecular Theory of Gas

Kinetic Energy – energy in motion

- Postulates:
  - Gas is a collection of particles that travel randomly along a straight-line path.
  - Molecules do not occupy a definite volume.
  - Molecules exhibit perfectly elastic collision.
  - Molecules do not exhibit attraction nor repulsion.

Gas Laws

- **Boyle's Law:**
  - At constant temperature, the pressure exerted by a gas is inversely proportional to its volume.
  - The graph of pressure against volume is hyperbolic.
  - When volume is doubled, the pressure is reduced to one-half.
  - P1V1 = P2V2

- **Charles' Law**
  - At constant pressure, the volume of a gas is directly proportional to its temperature expressed in Kelvin.
  - The graph of volume against the temperature is a straight diagonal line. Increasing the temperature (K) twice will double the volume of the gas.
  - V1T2 = V2T1

- **Gay-Lussac's Law**
  - At constant volume, the pressure exerted by gas is directly proportional to the temperature in Kelvin.
    - The graph of pressure against the temperature is a straight diagonal line.
    - Increasing the temperature twice will double the pressure exerted.
    - P1T2 = P2T1

- **Combined Gas Law**
  - P1V1T2 = P2V2T1

- **Avogadro's Law**
  - The amount of gas is directly proportional to its volume.
  - A mole of gas at standard temperature and pressure (STP), at 0oC and 1 atmosphere, occupies a volume of 22.4 liters (L).
  - When the amount is doubled, the volume is also doubled.

- **Ideal Gas Law**
  - PV = nRT where: P – pressure; V – volume; n – number of moles; T temperature in Kelvin; R is the gas constant: 8.3145 J/mol-K

Atomic Structure

- **Atom**
  - Building blocks of matter
  - Composed of 3 particles
    - **Protons** – positively charged; located in the nucleus
    - **Electrons** – negatively charged; orbiting around the nucleus
    - **Neutrons** – no charge particle located in the nucleus
  - Has equal number of protons and electrons
  - **Atomic number** is the same as the number of protons, while the **atomic mass** is the sum of the number of protons and neutrons
  - Exists in different versions called **isotopes**

    Ex. Isotopes of hydrogen : protium (H-1) = 1 proton, 1 electron, 0 neutron
    Deuterium (H-2) = 1 proton, 1 electron, 1 neutron
    Tritium (H-3) = 1 proton, 1 electron, 2 neutron

- **Formula Mass** – mass of an ionic compound

  Ex. $Ba(OH)_2$: Ba =(1 Ba atom)(137.3 g/Na atom) = 137.3 g
  O = (2 O atoms)(16 g/O atom) = 32.0 g
  H = (2 H atoms)(1 g/H atom) = 2.0 g
  Formula mass = 171.3 g

- **Molar Mass** – mass of one mole of a substance, which is equivalent to the formula mass expressed in grams

  The molar mass of the Ba(OH)2in the above example is 171.3g/mol.

- **Percentage Composition of a Compound**

  Ex. Find the % composition of $Ba(OH)_2$
  Ba : (137.3/ 171.3) (100%) = 80.2%
  O : (32.0/171.3) (100%) = 18.7%
  H : (2.0/171.3) (100%) = 1.2%
  Percent composition = 100%

- **Chemical Bonds** – forces that join atoms together to form molecules
  - **Ionic Bond** – formed between charged particles called ions due to electrons transfer

    Ionic bond forms the NaCl compound.
  - **Covalent Bonds** – formed when atoms share electrons
  - **Non-Polar Covalent** – bond between atoms of the same kind that share electrons equally.
  - **Polar Covalent** – bonds between atoms of molecules that do not equally share the electrons
  - **Metallic Bonds** – bonds between metals

EARTH SCIENCE

- Branch of physical science that deals with the physical aspects of the Earth
- The study of Earth Science encompasses the following topics:
  - Earth's origin and its relation to other celestial bodies
  - Earth's structure and composition
  - Earth's activities that bring about changes in the condition of the planet

Origin of the Earth

- The "Big-Bang Theory" is the closest in explaining how the Earth evolved.
- In this theory, the Earth started to exist at the time the solar system originated.
- The other theories that may explain the origin of the Earth are the following:
  - Planetissimal Theory
  - Binary Stars Theory
  - Dust-Cloud Theory

The Solar System

- Composed of several celestial bodies, namely:
  - The sun, which is the central figure of the system
  - The planets, which are generally characterized by its revolution around the sun
  - The moons or natural satellites that are revolving around its mother planet
  - The asteroids, meteors, meteorites, and comets
- The solar system is composed of inner planets: Mercury, Venus, Earth, and Mars. The outer planets are Jupiter, Saturn, Uranus, and Neptune.
- Pluto, which was once an outer planet, was temporarily excluded in the solar system because astronomers were not able to track the movement of the planet.

Our Moon: its Effects on the Earth

- Lone natural satellite of the Earth
- It makes one complete revolution in 27 1/3 days but its apparent revolution, that is, completing its phases, takes 29 ½ days.
- One effect of the moon is the occurrence of eclipse. **Solar eclipse** occurs when the moon passes directly between the Earth and the sun at each new moon.
- **Lunar eclipse** occurs when the moon passes through the shadow of the earth, and this happens during full moon.
- Another effect of the moon is the occurrence of tides on the Earth. **Spring tides** occur when the moon and the sun are aligned every two weeks (new moon and full moon) resulting in high tides, which are higher than usual.
- **Neap tides** occur when the sun and the moon are at right angles to each other (first and last quarters), resulting in high tides, which are lower than usual.

The Motions of the Earth and their Effects

There are two principal motions of the Earth:

- **Rotation**
  - The Earth rotates on its axis every 23 hours, 56 minutes, and 4.09 seconds. This is the length of the day on Earth.
  - As the Earth rotates, half of the Earth is lighted by the sun. This part experiences daytime. The other side which does not receive light experiences night time.
- **Revolution**
  - The Earth revolves around the sun in 365 ¼ days. This is the length of the day.
  - The seasonal change is caused by the tilt of the Earth and the angle of sunlight where it hits the Earth.

The Four Spheres (Layers) of the Earth

- **Lithosphere**
  - It is the solid, rocky crust covering the entire planet. It is made up of the following layers that extend from the outermost to the innermost part of the Earth: crust, mantle, outer core, and inner core.
- **Hydrosphere**
  - It is composed of all the water on or near the surface of the Earth.
  - This includes the oceans, lakes, rivers, streams, and even the moisture in the air.
- **Atmosphere**
  - It is the body of air that envelops the Earth.
  - The atmosphere is composed of 80% nitrogen and just around 16% oxygen. The small amount remaining, about 4%, is made up of other gases, such as carbon dioxide, helium, argon, and neon.
- **Biosphere**
  - The biological sphere came from "bio," which means life.
  - It is composed of all living organisms – the plants, animals, and other organisms.

Composition of the Earth

The Earth is composed of the materials and substances found in the lithosphere, hydrosphere, and atmosphere, which are solid materials, water, and air, respectively. Of these three, the solid materials are predominantly visible and diversified. These are the minerals, rocks, and soil.

- **Minerals**
  - **General Characteristics**
    - Single elements or compounds found naturally in the Earth's crust
    - Inorganic, that is, not derived from living things
    - Solids
  - **Visible Characteristics**
    - **Color** – the most dominant color of the mineral

- **Streak** – the color of a thin layer of a mineral. This is sometimes seen as a line of color in a mineral.
- **Luster** – the ability of a mineral to reflect, refract, or absorb light falling on its surface
- **Crystal form** – the geometric pattern of the mineral due to the arrangement of atoms and molecules
- **Cleavage and fracture** – definite flat surfaces that yielded after the breaking of minerals

▪ **Distinguishing Properties**
- **Hardness** – the ability of a mineral to scratch or cut another mineral
- **Specific Gravity** – density of the substance compared to the density of water

▪ **Classification of Minerals**
- **Silicious** – ex. Quartz, Agate, Hornblende
- **Non-metals** – ex. Calcite, Sulfur, Gypsum
- **Metal-ore** – ex. Gold, Silver, Copper
- **Gems** – ex. Jade, Topaz, Opal

- **Rocks**

The study of rocks is called Petrology. They are classified according to their color, texture, composition, and origin. Based on their origin, rocks are classified as:

▪ **Igneous rocks**
- Formed as magma cools off and crystallizes
- Those formed inside the Earth are called intrusive, while those outside are called extrusive rocks.
- Examples are volcanic tuff, granite, and basalt.

▪ **Sedimentary rocks**
- Formed from sediments of weathered materials
- Used to tell much of Earth's history because these rocks contain fossils
- Examples of these rocks are limestone, gravel, and pebbles.

▪ **Metamorphic rocks**
- These rocks undergo stages of development due to heat, pressure, and chemical reaction.
- Examples of these rocks are slate, marble, and graphite.

- **Soil**

Soil is a natural resource formed out of weathered rocks with humus on the Earth's surface. It consists of visually and texturally distinct layers, which are classified as:

▪ **Topsoil** – the uppermost layer with decomposed organic matter, mixed with small amount of minerals. Plants mostly grow in topsoil because it is rich in humus.

▪ **Subsoil** – This is also called mineral layers because it is composed of the soil and its parent mineral. This contains less humus and also acts as reservoir of water.

- **Soil Conservation**

The following are some of the ways to conserve soil:

- **Contour plowing** – plowing in furrows which follow the contours of the land
- **Strip cropping** – planting crops arranged in alternate bands of row crops and cover crops
- **Terracing** – construction of steplike ridges following the contours of the slopes of the land
- **Crop rotation** – alternate planting of row crops one year with cover crops the next year

Activities that Change the Condition of the Earth

- **Weathering** – the breaking down of rocks by physical or chemical means
- **Erosion** – the transport of weathered materials
- **Diastrophism** – the movement of the landforms
- **Continental drift** – a theory that explains the drifting of one continent to its present position
- **Plate tectonics** – This theory explains that the outermost rigid layer of the Earth, called plates, is continuously moving. This also explains the cause of earthquakes, volcanoes, ocean trenches, and formation of mountains.
- **Rock deformation** – refers to the changes in volume and shape of rocks that may cause fractures and lead to the formation of faults
- **Earthquake** – vibration resulting from the movement along the existing fault line. The point where the vibration originated is called **focus**. The point on the surface which is directly above the focus is known as **epicenter**.
- **Volcanism** – covers all kinds of volcanic activities, which include formation of volcanoes, building, and expulsion of magma. Volcanoes may be classified according to cones:
  - **Cinder cone** – with narrow base and steeper slope
  - **Shield cone** – with broad base and gentle slope
  - **Composite cone** – with a nearly perfect shape whose slope is gentle at the base and become steeper as one approaches the peak.

Weather and Climate

- **Meteorology** is the study of weather and climate.
- **Weather** refers to the atmospheric condition at a particular place and time.
- **Climate** is the pattern of weather in a bigger land area over a long period of time.
- **Tropical disturbances** are classified as tropical depression, storm, and typhoon.

These disturbances are measured by the strength of accompanying winds.

- The prevailing winds in the Philippines are the northeast monsoon and the southwest monsoon.

BIOLOGICAL SCIENCE

BIOLOGY

- Coined from the Greek words 'bios' meaning life, and 'logos' meaning study of
- Study of the living matter both plants and animals and the plant-like and animal-like organisms.
- Branches and fields of study
  - **Anatomy** – macroscopic structure of multi-cellular organisms
  - **Biochemistry** – chemicals of life, like carbohydrates, proteins, amino acids
  - **Botany** – structure, characteristics, classification of plants
  - **Cytology** – properties and structure of the cell
  - **Embryology** – embryo or developing organism
  - **Entomology** – kinds, characteristics, and structure of insects
  - **Ecology** – interaction between the organism and its environment
  - **Genetics** – heredity or transmission of traits from parents to offspring
  - **Herpetology** – reptiles and amphibians
  - **Ichthyology** - fish
  - **Molecular biology** – structure and function of molecules in the living things
  - **Morphology** – form and structure of living organisms
  - **Microbiology** – structure of microorganisms and their effects on other organisms
  - **Marine biology** – marine organisms
  - **Ornithology** – birds
  - **Parasitology** – parasites
  - **Pathology** – disease-causing organisms
  - **Paleontology** – fossils
  - **Physiology** – physical function of living organisms
  - **Zoology** – animals

Living Things

- Composed of cells
  - Unicellular (1 cell) or multicellular (many cells)
- Exhibit levels of organization: complex and organized
  - Cell – tissue – organ – system – organism
  - Cell as the basic unit of structure
- Grow and develop
  - Increase in size (grow) develop (less to more complex organism) through cell division

    Mitosis – division in somatic or body cells

    Meiosis – division of gametes or sex cells
- Respond to stimulus/stimuli
  - Ability to react to any change in condition

    Stimulus – the condition to which an organism reacts/responds

- Need and Transform energy
  - Energy is required in order for the organism to grow, develop, and later reproduce.
- Adapt or adjust
  - Ability to adjust to conditions in its environment in order to survive

    Tropism: phototropism (light); hydrotropism (water); geotropism (gravity)

    Mimicry, camouflage, growing thorns, producing chemicals that affect the other organisms
- Reproduce – the ability to produce a new organism of its own kind
  - **Asexual** – no sex cells involved; one parent only: binary fission, budding, spore formation, regeneration, vegetative propagation
  - **Sexual** – sex cells or gametes are involved; male and female parents

Microscope

- A tool necessary in the study of the organisms
- Composed of lenses used to produce image of the organism under study. The images are magnified many times so the eyes can see.
- Come in different types:
  - **Light Microscope** – refracts light using lenses to magnify the image formed. Its usual magnification: 40x , 100x, 400x
  - **Stereoscope** – 3D view, 10-20x magnification, viewing with 2 eyes
  - **Scanning Electron Microscope** – uses beam of electrons to magnify an specimen two million times
  - **Transmission Electron Microscope** – uses beam of electrons that pass through the specimen allowing to view the inner parts of the specimen

Cell Theory

- Describes or explains what a cell is.
  - The cell is the basic component of a living organism.
  - New cells are produced from existing cells.
  - Cell is the building block of life.
- Kinds:
  - **Prokaryotic** – has no nucleus; ex. Bacteria
  - **Eukaryotic** – has true nucleus; ex. Mammals
- Parts / Structures
  - **Cell Wall** – protects and supports the plant cell; allows entry of water and other substances into the cell
  - **Cell membrane** – between the cell and its environment which maintains homeostasis
  - **Nuclear membrane** – controls that substances entering and leaving the nucleus
  - **Nucleus** – carries the genetic material and controls cellular activities
  - **Cytoplasm** – jelly-like part outside the nucleus that houses and protects the organelles
    - **Endoplasmic Reticulum** – network of tubes that carries materials through the cell

- Ribosome – synthesize proteins
- **Mitochondrion** – breaks down sugar and releases energy; where aerobic respiration takes place
- **Vacuole** – food or water storage
- **Lysosome** – breaks large food molecules to smaller ones
- **Chloroplast** – where photosynthesis takes place
- **Nucleolus** – 'nucleus' inside the nucleus responsible for making ribosomes
- **Golgi Apparatus** – modify and export proteins
- **Centrioles** – separate chromosome pairs during mitosis

- **Transport of Materials through Cell Membrane or Plasma Membrane**

Whether or not a substance can pass through the plasma membrane is determined by the permeability of the membrane.

- **Permeable** – Almost all substances can pass through.
- **Impermeable** – No substance can pass through.
- **Selectively permeable** – Some substances can pass through, while others cannot.
- **Active Transport** – Energy is required so substances can pass through.
- **Passive Transport** – Substances can pass through without energy use.
  - **Diffusion** – Substance moves or spreads from area of higher concentration to an area of lower concentration
  - **Osmosis** – Spreading of water molecules from region of higher concentration of solute to region of lower concentration.

    Osmotic Pressure – force exerted by the water molecules that spread through the area
    - **Hypotonic** – less solute concentration in the area, so water moves away
    - **Hypertonic** – greater solute concentration in the area, so water moves toward
    - **Isotonic** – equal concentration on both sides of membrane, so water does not diffuse
- **Endocytosis** – engulfing materials
  - **Phagocytosis** – Particles are engulfed in the vacuole.
  - **Pinocytosis** – Fluid is enclosed.

Energy: Requirement for Survival

- Energy is a requirement for an organism to survive. It enables one to do work. The energy needed by an organism is basically coming from the food it takes. The food also has taken its energy from the sun.
- **Classification of organisms based on energy**
  - **Producer / autotrophs** – makes its own food through the process of photosynthesis

    During photosynthesis, carbon dioxide and water are converted to carbohydrates with the use of sunlight as energy source. The process releases oxygen.

    $$6CO_2 + 6H_2O \longrightarrow C6H_{12}O6 + 6O_2$$
  - **Consumers/Heterotrophs** – obtain energy from other organisms
    - First-order consumer / herbivores

- Second-order consumer
- Third-order consumer

- **Decomposers/Saphrotrophs** – special type of consumers because they obtain energy by consuming both the producers and consumers when they are already dead.

Cell Division

- **Mitosis** – Body cells (somatic cells) are produced for growth and development.
- **Meiosis** – Gametes or sex cells are produced, and these are necessary for reproduction.
- **Cell Cycle** – exhibits the stages in the life of a cell

  G1 ⟶ S ⟶ G2 ⟶ M

  - **Synthesis / Replication** (S) – the making of the exact copy of the genetic material so the daughter cells will have the same genetic make-up as the parent cell
  - **Gaps** (G1 and G2) separate DNA Synthesis from Mitosis; mediation to switch in cellular activity
  - **Mitosis** (M) – separation of the copied genetic materials into the daughter cells

- **Stages of Mitosis**
  - **Interphase** – resting stage; includes the G1, S, and G2
  - **Prophase** – chromosomes become visible and condensed; nuclear membrane disappears; centrioles move toward opposite poles; spindle fibres from centrioles and attach to chromatids
  - **Metaphase** – centrioles reach the poles; chromosomes line up in the equator.
  - **Anaphase** – spindles shorten, pulling the sister chromatids apart.
  - **Telophase** – chromosomes become less condensed; nuclear envelope forms; cytokinesis happens.

- **Cytokinesis** – breaking or division of the cytoplasm to form the daughter cells

- **Meiosis** – reduces the chromosome by one-half (diploid cell becomes haploid)
  - Haploid daughter cells become diploid after fertilization.
  - Half of the chromosomes come from the female gamete, while half comes from the male gamete.

- **Cancer** – develops when the cells divide uncontrollably

1. Which of the following steps is not a part of the scientific method?

   A. Interview resource persons in order to collect possible data
   B. Make a guess about the answer to the problem and predict what will happen
   C. Identify and state the problem
   D. Perform the experiment to test prediction

   Answer – A: Interviewing resource persons is not a part of the basic steps in scientific method, although it can be used in some scientific studies. B is formulation of hypothesis, C is about the problems, and D is experimentation. These three are included in the basic steps in solving problems.

2. Which of the following statements does a scientist always consider?

   A. Accurate observations are necessary to organize problems.
   B. Predictions do not point the way to possible solutions to problems.
   C. Graphs are not useful in interpreting data.
   D. Experiments provide information that will always support predictions.

   Answer – D: Experimentation is the most important step in the scientific method. The results of the experiment or the information provided by the experiment will the answer the problem and will prove or disprove the hypothesis.

3. What is the importance of having a control set-up in any scientific investigation?

   A. It will identify the variables in the experiment.
   B. It will provide a basis of comparison between treated and untreated subjects.
   C. It will provide a basis for measuring data.
   D. It will eliminate mechanical errors in experimentation.

   Answer – B: A control set-up is one where the manipulated variables or factors in an experiment are changed or removed. For this matter, the results in the control set-up serve as the basis of comparison between the treated and untreated subjects.

4. Which among these measurements is the comparison between the quantity to be measured and its corresponding standard in measuring length?

   A. Meter
   B. Mass
   C. Ruler
   D. Liter

   Answer – A: Length is one of the fundamental quantities. The fundamental unit of measuring length is the meter and the standard device used in measuring length is the meterstick.

5. Of the following situations, which one is not accelerated?

   A. A ball rolling down an inclined plane
   B. A car traveling at constant velocity along the expressway
   C. An object which is whirled horizontally in uniform circular motion
   D. A stone which is freely-falling

Answer – B: A body is said to be accelerated if there is a change in velocity. In B, the car which is moving at constant velocity has no change in velocity; thus, is not accelerated. A and D are accelerated because the bodies move faster as they move downward. In C, even if the object is in uniform circular motion, its velocity changes because at any point in circular path, the direction of the velocity changes so the velocity being a vector changes. Thus, the body is accelerated.

6. Which of the following is true when a block of wood lays at rest on top of the table?

   A. There are no forces acting on the block.
   B. The sum of the forces acting on the block is zero.
   C. The block is in uniform motion.
   D. The block is accelerated.

Answer – B: There are forces acting on the block; the weight of the block and the upward force exerted by the tabletop on the block. However, the two forces acting on the block are equal and opposite in direction (third law of motion), the sum of these forces is zero. Therefore, there is no net force, and so, it is at rest.

7. A rectangular block of iron has a dimension of 5 cm x 10 cm x 15 cm and weighs approximately 60 N. How should this block be placed on top of the table so as to exert the greatest pressure on the surface?

   A. On the 5 cm x 10 cm side
   B. On the 5 cm x 15 cm side
   C. On the 10 cm x 15 cm side
   D. On any of the sides

Answer – A: Pressure is force exerted in a given unit area, $p = F/A$. The given equation shows that the smaller the area, the greater is the pressure. Since the 5 cm x 10 cm side has the smallest area, it will give the greatest pressure when placed on this side.

8. A razor blade can float on the surface of water because of ___________.

   A. Surface tension
   B. Capillarity
   C. Buoyant force
   D. Liquid pressure

Answer – A: A razor blade can be made to float on the surface of water even though it is denser in water because of surface tension. The surface of the water acts like it is under tension due to the attractive forces between molecules, creating a mat-like structure on the surface of the water. This prevents the razor blade to sink in water but float instead.

9. The following sources of energy are considered renewable and sustainable, except ___________.

   A. Sun
   B. Waterfalls
   C. Wind
   D. Coal

Answer – D: Coal cannot be replenished nor recycled. Continuous use of this source for many years may result in total loss of this form. Unlike the sun (solar), waterfalls and wind are classified as renewable, sustainable, and even inexhaustible when properly developed.

10. Of the following sources of energy listed below, which one is caused by the concentration of heat in the Earth's crust and gives detrimental effect on the population's food supply?

A. Solar
B. Geothermal
C. Wind
D. Nuclear

Answer – B: Geothermal energy is caused by the concentration of heat in the earth's crust, and this heat comes to the surface as natural steam by way of hot springs, geysers, or other steam vents. This source gives detrimental effect on the food supply because it affects primarily the vegetation of the surface of the land.

11. Which of the following statements is true?

A. Light travels fastest in a vacuum, while sound travels fastest in solid.
B. Light travels fastest in solid, while sound travels fastest in liquid.
C. Both light and sound travel fastest in a vacuum.
D. Both light and sound travel fastest in solid.

Answer – A: Light travels the fastest in a vacuum and the slowest in solid. Conversely, sound travels the fastest in solid and the slowest in air. Sound needs a medium to be transmitted, and the best medium for sound transmission is solid. Light, meanwhile, does not involve any medium and can even travel without a medium.

12. Of the following, which is one is considered an alkaloid?

A. Aspirin
B. Iodized salt
C. Muriatic acid
D. Baking soda

Answer – D: Alkaloid is a base and baking soda is a basic substance. Aspirin and muriatic acid are acidic, and iodized salt is an example of salt.

13. Perfumes contain organic compounds called ____________.

A. Esters
B. Ethers
C. Acids
D. Bases

Answer – A: Esters are organic compounds made from a carboxylic acid and an alcohol. The odors of the simple esters; are generally regarded as very pleasant. Many of the odors of fruits and flowers result from mixtures of esters; that is why, many of them whether natural or synthetic are used in perfumes and food flavorings.

14. The chemical equation $AB + \text{Energy} \rightarrow A + B$ is an example of ____________.

A. Exothermic decomposition
B. Exothermic synthesis
C. Endothermic decomposition
D. Endothermic synthesis

Answer – C: The equation shows that energy (usually heat) is added or absorbed resulting to the decomposition (A + B) of the compound AB. The chemical reaction is endothermic decomposition.

15. One of the properties of gases is its compressibility. This is because the molecules are ____________.

A. Close together
B. Far apart
C. Stationary
D. Always moving

Answer – B: The molecules of gases are far apart, and so, with enough spaces to be filled out during compression. Of the three states of matter, gases are highly compressible, followed by liquid and then solid, and these are due to the arrangement of matter.

16. Which of the following is true of metalloids?

A. They conduct heat and electricity more effectively than metals.
B. They both react with non-metals.
C. They have the properties of both metals and non-metals.
D. They conduct electricity better than non-metals.

Answer – C: Metalloids share the properties of both metals and non-metals. For example, a metalloid may be shiny, which is a property of a metal, and at the same time brittle, which is a property of a non-metal. Examples of metalloids are arsenic, antimony, and silicon.

17. Which of the following substances listed below is considered a colloid?

A. Water
B. Alcohol
C. Soda
D. Milk

Answer – D: A colloid is a substance whose particles measure between 1 to 100 nanometers (1nm = 10-9 m). It serves as the link between solution and suspension. Unlike suspension, a colloid does not settle, and passes through a filter paper. That is why, it is considered as homogenous mixture. An example of a colloid is milk.

18. Sterling silver which is an alloy of copper in silver is considered to be a ____________.

A. Solid-solid solution
B. Solid-liquid solution
C. Solid-gas solution
D. Liquid-liquid solution

Answer – A: Sterling silver is an alloy of two metals, copper, and silver; thus, it is considered as a solid-solid solution.

19. Which of the following is an extensive property of matter?

A. Weight
B. Melting point
C. Specific gravity
D. Color

Answer – A: Extensive property depends on the amount of material, while intensive property depends on the quality of the material. Of the four choices, weight is an extensive property of matter because it depends on the amount of force exerted by the Earth on that matter.

20. The astronomer who developed the theory that the sun is the center of the solar system was ___________.

A. Newton
B. Einstein
C. Ptolemy
D. Copernicus

Answer – D: It was in the sixteenth century when Nicolaus Copernicus, a Polish astronomer, enunciated the principle of heliocentric planetary motion, stating that the sun, and not the Earth, was the center around which the planets revolved. This principle rejected the idea of Ptolemy (Ptolemeic Theory) where the Earth is the center around which the other celestial bodies, including the sun revolved.

21. The Earth's rotation on its axis gave rise to the occurrence of day and night. What would have happened if the Earth's axis is not tilted?

A. Some places would just have day time, while others only night time.
B. Some places would have very long nights; others would have short days.
C. The days and nights throughout the globe would always be of equal length.
D. The length of day and night would just be the same as when the axis is tilted.

Answer – C: Since the Earth is tilted as it rotates on its axis, longer daytime occurs in the northern hemisphere during summer and longer night time during winter. If the earth's axis is not tilted, all the parts of the Earth will be receiving equal amount of light from the sun, and so, the days and nights throughout the globe would always be of equal length.

22. How many phases does the moon pass during one complete revolution around the Earth?

A. 3
B. 4
C. 6
D. 8

Answer – D: During one apparent revolution around the Earth (291/2 days), the moon passes through eight various phases from complete darkness to complete light and back to complete darkness. These phases are: new moon, waxing crescent, first quarter, waxing gibbous, full moon, waning gibbous, last quarter, waning crescent, and back again to new moon to start the cycle. Waxing means the phase is becoming larger, and waning means the phase is becoming smaller.

23. What planet is closest to the Earth and with a distance of about 42 million kilometers?

A. Mars
B. Venus
C. Mercury
D. Jupiter

Answer – A: Of the three neighbor planets of the Earth, Mars is the one closest to the Earth. But even at its closest, Mars is still 35 million miles or around 42 million kilometers from Earth.

24. What term is given to the surface of the Earth between the tropic of cancer and the equator?

A. Great circle
B. Meridian
C. Zone
D. Solstice

Answer – C: The region of the surface of the Earth that are divided according to prevailing climate and latitude is known as zone. In some cases, a zone can be identified as the surface of the Earth between two adjacent great circles.

25. The greatest number of storm casualties during the Super Typhoon Yolanda was caused by _______.

A. Strong winds
B. High rainfall
C. Storm surge
D. Low pressure

Answer – C: A storm surge is a flood of ocean or sea water that accompanies the typhoon. It is considered as the most devastating feature of a typhoon that strikes coastal area. This happened in Tacloban, Leyte during the height of Super Typhoon Yolanda, where it gave the greatest number of casualties in the history of typhoon events.

26. Prokaryotic cells are different from eukaryotic cells because prokaryotes have _______.

A. Small ribosomes used for protein synthesis.
B. Nucleic acids found inside the nucleus.
C. Mitochondria that participate in cellular respiration.
D. Genome that is located in the nucleoid.

Answer – A: Prokaryotic cells or prokaryotes are cells without true nucleus, while eukaryotic cells have true nucleus. Prokaryotes do not have a membrane-bound organelles like mitochondria. Since prokaryotes have no true nucleus, the nucleic acids and the genome are not evident in these cells. However, prokaryotes have small ribosomes that are used in protein synthesis.

27. What type of fungus is used in making alcoholic drinks and help prepare a dough in baking?

A. Mold
B. Yeast
C. Mushroom
D. Algae

Answer – B: Yeast is a type of fungus and classified under Ascomycota or sac fungi. Yeast is important in the fermentation process of making alcoholic drinks like wine and liquor. It is also involved in leavening the dough during baking process.

28. In order to determine the cause of infectious disease, a diagnostic test that allow microorganisms to multiply in a medium is performed in a controlled laboratory condition. This test is called ___________.

A. Complete blood count (CBC)
B. Urinalysis
C. Microbial culture
D. Hemoscope

Answer – C: Microbial culture is the growing of microorganisms, such as bacteria in a nutrient medium, which is usually agar, for the purpose of determining the cause of infectious disease or other ailments. The culture is done in a controlled laboratory so as not to be contaminated by other microorganisms.

29. A two-lobed endocrine gland, shaped like a bow tie, which is located at the neck and regulates metabolism is known as ___________.

A. Pituitary gland
B. Thyroid gland
C. Parathyroid gland
D. Adrenal gland

Answer – B: Thyroid gland is one of the endocrine or ductless glands which is located at the neck. It secretes hormone called thyroxine that stimulates metabolic rates.

30. A woman cannot bear a child because of a defective fallopian tube. However, a modern technological process could help her bear a child through the union of ovum and sperm inside a test tube. This process is called ___________.

A. Test tube
B. Tubal ligation
C. In vitro fertilization
D. In vivo fertilization

Answer – C: Fertilization in woman occurs in the fallopian tube, so if the fallopian tube is defective, no fertilization will happen, and the woman will not bear a child. However, a modern technological process could help the woman bear a child by getting matured egg from the wife and sperm from the husband, and the union of this ovum and sperm takes place outside the organism (in vitro means outside), and usually happens in the test tube. That coins the term "test tube baby".

31. Bamboos, which are considered as the tallest grass reproduce asexually through extended structures, are called ___________.

A. Roots
B. Buds
C. Runners
D. Seeds

Answer – C: Runners are slender, extended stems that put forth roots from nodes spaced at intervals along its length. Bamboos reproduce asexually by cutting these stems and staking them in soil.

32. Dengue is an infectious disease transmitted by a carrier. This carrier is classified under phylum arthropoda and Class ___________.

A. Arachnida
B. Chordate
C. Insecta
D. Monera

Answer – C: Dengue is transmitted by a mosquito known as *Aedis aegipte.* Mosquito is classified under *phylum arthropoda* (joint legged) and class *insecta,* which is mainly characterized by having three pairs of legs.

33. The branch of biology that deals with the study of animal behavior is called ________.

A. Psychology
B. Taxonomy
C. Ecology
D. Ethology

Answer – D: While Psychology deals with human behavior, Ethology is a branch of biological science that deals with animal behavior.

34. Decomposition is a process in which a substance is broken to make it useful in other processes. The organisms that are capable of decomposing a substance are called ________.

A. Autotrophs
B. Heterotrophs
C. Saprotrophs
D. Producers

Answer – C: Saprotrophs came from two Greek words, *sapros*, which means decaying *matter*, and *trophe*, which means nutrients or nourishments. This means that the food nutrients of these organisms are coming from the decaying matter.

35. Which of the following is true of recessive genes?

A. They are more aggressive and have more superior phenotypic traits.
B. They will only have phenotypic expression if they are homozygous.
C. They will prevent a dominant gene from expressing its phenotype.
D. They should be paired with dominant gene in order to be expressed.

Answer – B: Recessive traits are governed by recessive genes. A recessive trait appears in an individual only if the dominant gene governing the characteristic is not present. Organisms must be homozygous recessive; that is, must contain two recessive genes before the recessive trait is seen in an individual. If a recessive gene is masked by a dominant one, then, it is still dominant, meaning, a dominant gene can be expressed even if one copy of the gene is present.

36. Which of the following describes the habitat of an organism?

B

A. It is the population of an ecosystem.
B. It is the food supply in a given environment.
C. It is a place in the environment where the organisms live.
D. It is a community of an ecosystem.

Answer – C: Habitat is a place or a certain area in an environment, which an organism or a biological population lives or occurs. A pond if considered as a place is an example of habitat.

37. Which of the following pairs of body parts is considered homologous?

A. Human Arms and Seal's Flippers
B. Appendix and Tail bone
C. Lungs and Gills
D. Ectoderm and Endoderm

Answer – A: A pair of body parts is considered as homologous if they are corresponding in structure and evolutionary origin. Human arms and Seal Flippers are examples of homologous parts. Lungs and Gills are analogous because they are similar in function but not in structure.

38. This segmented worm is used in the traditional medicine to facilitate anticoagulation. What is this organism?

A. Liver fluke
B. Leech
C. Earthworm
D. bloodworm

Answer – B: Leech is an aquatic bloodsucking segmented worm (*annelid*) which was formerly used by physicians to bleed their patients, thus facilitating anticoagulation of blood.

39. What hormones in plants are responsible in promoting auxillary budding and apical dominance?

A. Auxins and Gibberelins
B. Cytokinins and Gibberilins
C. Auxins and Abscisates
D. Cytokinins and Auxins

Answer – A: Auxins and Gibberilins are plant hormones that regulate and influence various developmental processes, such as root initiation, stem elongation, flowering, budding, apical dominance, and the like. Malunggay is a typical example of a plant with enough auxins and gibberilins.

40. During what part of the day is carbon dioxide released in plants?

A. Morning
B. Noon
C. Afternoon
D. Evening

Answer – D : One of the characteristics of organisms is the ability to respire and since plants are organisms, they undergo respiration; that is, plants take in oxygen and give off carbon dioxide. However, the release of carbon dioxide in plants happens in the evening because carbon dioxide is needed by them during daytime in photosynthesis.

41. What do you call a marine mammal that has a long ivory tusks and well-adapted in the tundra biome?

A. Walrus
B. Whale
C. Polar bear
D. Elephant

Answer – A: Walrus is a marine mammal (not a whale) that has a long ivory tusks (not an elephant) and well adapted in the tundra biome because of its thick skin. Polar bear is also adapted in the tundra not because of its thick skin but because of its thick hair.

42. What are the two chemical factors that cause decomposition?

A. Putrefaction and respiration
B. Putrefaction and mitosis
C. Autolysis and putrefaction
D. Autolysis and photosynthesis

Answer – C: Autolysis is the decomposition of tissues or cells of an organism by autogenous substances, such as enzymes. Putrefaction, on the other hand, is decomposition of organic matter by microorganisms.

43. Commensalism is an interaction between two organisms, whereby one organism is benefited, while the other is not affected. This is exemplified by which of the following?

A. Two spiders fight each other for survival.
B. A flea feeds on the blood of the dog.
C. A bird following the army ant is able to catch and eat the fleeing insects which are caused by the army ant colony.
D. A soaring eagle catches and devour the chick.

Answer – C: The bird and the army ant interact by commensalism. The bird is able to catch and eat the insects caused by the colony of the ant. In this case, the bird is benefited because it feeds on the insects, while the ant is not affected at all. Choice A is competition, Choice B is parasitism, and Choice D is predation.

44. The interrelation between the physical and biological components of the environment is called __________.

A. Ecosystem
B. Ecological niche
C. Water system
D. solar system

Answer – A: Ecosystem is not a habitat nor a group of organisms but rather a condition, whereby the physical (abiotic) and the biological (biotic) components of the environment are interrelated.

45. A forest suffered from wildfires; however, the remains are still considered to be viable. Five years later, trees and other plants are seen growing in this forest. This condition is called __________.

A. Primary ecological succession
B. Secondary ecological succession
C. Tertiary ecological succession
D. Quarterly ecological succession

Answer – B: Ecological succession is the natural recovery or restoration of a damaged ecosystem. Basically, there are two types of ecological succession: primary and secondary succession. In secondary succession, there are remnants or remains of the previous ecological community that may develop into another community, succeeding the previous one. In primary succession, there are no remnants of the previous community. Instead, there is initial establishment or development of an ecosystem.

46. The term used in showing the feeding connections among all the living things is called ________.

A. Energy pyramid
B. Water cycle
C. Nutrition cycle
D. Food web

Answer – D: There are two kinds of feeding connections or food transfer: food chain and food web. In food chain, there is only a single path of food transfer. In food web, the transfer of food branches out in different directions creating a web-like structure of who eats whom.

47. A simple hypothetical ecosystem consists of the following: worms, which are the primary consumers feed on the cabbage, the producers. The small passerine birds, feeding on the worms are the secondary consumers, and the lion acts as the tertiary consumer. A farmer arrives and sprays pesticides, killing all the caterpillars. What will happen to the organisms in the ecosystem?

A. The lion will die.
B. The passerine birds will thrive.
C. The cabbage will wilt and die.
D. The worms will still reproduce.

Answer – A: The hypothetical situation shows that the passerine bird feeds on the worm, so, when the worm dies, the bird dies also and so, eventually, the lion will die because the lion feeds on the passerine bird.

48. Red tide is caused by the increase in the number of planktons in the sea. What particular marine organism is affected by these planktons and causing detrimental effect on humans?

A. Fish
B. Mollusks
C. Arthropods
D. Corals

Answer – B: The most common marine organisms affected by the planktons (Dino flagellates) that caused red tide are the shell-fish, locally known "tahongs". Shell-fish is under Phylum *Molluska* or mollusks.

49. The major concern about global warming arises from increased concentration of ________.

A. Greenhouse gases
B. Acid rain
C. Photochemical smog
D. Aerosols

Answer – A: Global warming is a natural or human-induced increase in the average global temperature of the earth's atmosphere. This abnormal atmospheric condition is mainly caused by the greenhouse gases that trapped the heat energy radiating from the Earth's atmospheric system. This process of trapping heat is called greenhouse effect. The most common greenhouse gases include carbon dioxide, methane, and chlorofluorocarbons a (CFCs).

50. The Kyoto Protocol is one of the international assemblies whose advocacy is environmental protection. Thus, this assembly requires the countries to participate in this advocacy by reducing __________.

A. Greenhouse emission
B. Photosynthesis
C. Acid rain
D. Reforestation

Answer – A: The Kyoto Protocol was an assembly or conference participated by various leaders and environmentalists of some countries. This was held in Kyoto, Japan, thus, named Kyoto Protocol. Its advocacy is a lifetime environmental protection, and so, this assembly requires the participating countries, at least, to help in reducing greenhouse emission.